The Structural Radiante

A Pentadimensional Instrument for Visualization and Analysis of Structural Trajectories in Complex Systems

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Abstract
The Structural Radiante is the visualization and formal analysis layer of the Kappa Method (Ohio, 2026a). The Kappa Method detects structural instability in time series via five topological observables: the Ohio Number *Oh(t)*, structural memory *Φ(t)*, dynamic rigidity *η(t)*, structural diversity *Ξ(t)*, and state-phase deficit *DEF(t)*. The Radiante transforms these observables into a dimensional navigation instrument that answers the questions the Kappa raises but does not directly resolve: what regime is the system in? How fast is it moving toward rupture? What is the energetic cost of reorganization? Where does it go next? This paper presents: (i) the geometry of the pentadimensional space K(t) and the three archetypal regimes — Nagare (JP: *nagare*, flow), Utsuroi (JP: *utsuroi*, transition), and Katashi (JP: *katashi*, crystallization); (ii) the gravitational sub-observable Ψ(t), which quantifies the cost of structural reorganization as gravitational action; (iii) the domain-agnostic exchange schema radiante.v1; and (iv) empirical validations across four domains: the 2008 Global Financial Crisis via Kappa-FIN v3 (DOI: 10.5281/zenodo.18917558), the polarized political blogosphere (Adamic & Glance, 2005), educational dropout (OULAD dataset), and LLM hallucination detection via Kappa-LLM (DOI: 10.5281/zenodo.18883790). The GFC 2008 analysis demonstrates that the Katashi onset detected in November 2007 — ten months before the Lehman Brothers bankruptcy — constitutes the first retrospective validation case of the framework. Section 7 presents the multi-entity network extension that constitutes the architectural gateway for Kappa-GEO, an ongoing geopolitical application.

Keywords: topological data analysis; Kappa Method; regime transitions; complex systems; gravitational sub-observable; gravitational action; Kappa-FIN; Kappa-LLM; Kappa-GEO; Law of Ohio; persistent homology; Forman-Ricci curvature.

1. Introduction

The Kappa Method (Ohio, 2026a) is a mathematical framework for detecting structural instability in any system representable as an interaction network. Published under DOI 10.5281/zenodo.18883639, the Kappa Method has been validated in financial markets, large language models, and educational dropout prediction. The method delivers a five-component state vector for each temporal window: K(t) = (Oh, Φ, η, Ξ, DEF). The question the Radiante answers is: *how does one navigate this five-dimensional space?*

The etymology of the name is precise. In astronomy, the *radiant* is the point in the sky from which meteor trajectories appear to converge when projected onto 2D. The Structural Radiante performs the inverse operation — it takes high-dimensional trajectories and projects them into interpretable spaces, preserving the regime properties that matter for analysis. The instrument renders K(t) as an interactive 3D trajectory with configurable projection modes.

This paper also introduces the **gravitational sub-observable Ψ(t)** — the first formalization of the energetic cost of regime transitions in the Kappa ecosystem — and the **radiante.v1 schema**, a domain-agnostic

exchange format. Section 7 formalizes the multi-entity extension that serves as the architectural gateway for Kappa-GEO, the geopolitical extension of the Kappa Method currently in development.

The publication sequence — Kappa-FIN → Radiante → Kappa-GEO — is not merely chronological: it is **architectural**. Each layer presupposes the previous one. Kappa-GEO requires the gravitational formalism of Ψ and the multi-entity infrastructure of the `links` field, both of which are formally established here.

2. The Pentadimensional State Vector K(t)

2.1 Formal Definition

The pentadimensional state vector is defined as:

```
K(t) = (Oh(t), Φ(t), η(t), Ξ(t), DEF(t))
∈ [0,∞) x [0,∞) x [0,1] x [0,1] x [0,1]
```

Observable	Symbol	Interpretation in the Radiante
Ohio Number	Oh(t)	H1 homological pressure. Oh > 1 = supercritical regime — system above the irreversibility threshold
Structural Memory	Φ(t)	Integral of correlation anomalies. Accumulated topological damage without resolution
Dynamic Rigidity	η(t)	System resistance to reorganization. η → 1 = imminent gravitational locking
Structural Diversity	Ξ(t)	Entropy of topological paths. Ξ → 0 = channeling, loss of degrees of freedom
State-Phase Deficit	DEF(t)	Divergence between current state and expected phase space. Dynamic disconnection

The **structural viscosity** $\nu_s(t) = \tau_{Katashi} \times \eta_{Katashi} / (\Xi_{Katashi} + \varepsilon)$ is a derived observable that quantifies resistance to reorganization. In the GFC 2008 case: $\nu_s = 386 \times 1.123 / (0.286 + \varepsilon) = 1517.6$ — corresponding to complete gravitational locking before the September 2008 collapse.

2.2 Relationship with Kappa-FIN

Kappa-FIN v3 (Ohio, 2026b) delivers as output the file `kappa_v3_state.csv` with columns `Oh, phi, eta, Xi, DEF, regime` for each date. The Radiante data-adapter converts this CSV directly to radiante.v1 format, adding derived phases, regime classification, and the gravitational sub-observable. The demo `finance_2008_gfc_demo.json` in this repository is the Radiante visualization of real Kappa-FIN v3 results for the GFC 2008.

3. Structural Regimes

3.1 The Three Regimes

Regime	Japanese	Formal Criteria	Dynamics
Nagare	Nagare (JP: nagare, flow)	Ξ > 0.6, η < 0.4, DEF < 0.3	Adaptive flow — system distributes pressure across diverse topological paths
Utsuroi	Utsuroi (JP: utsuroi, transition)	Liminal state	Transition — maximum sensitivity to interventions; system left stability but not yet locked

Regime	Japanese	Formal Criteria	Dynamics
Katashi	Katashi (JP: katashi, crystallization)	$\eta > 0.6$, $\Phi > 0.6$, $\Xi < 0.4$	Crystallization — locked correlations, accumulated memory; precedes SPS rupture

3.2 Point of No Return (PNR)

The PNR is signaled when simultaneously: (i) Oh(t) > 0.85; (ii) regime curvature $\kappa(t) > 0.80$; and (iii) regime ≠ Nagare. In the GFC 2008, the formal irreversibility condition — $\Phi > \Phi^* = 0.184$ — was reached on 2007-11-13 with $\Phi = 0.241$. From this point, the system had no endogenous reorganization capacity. The Lehman Brothers bankruptcy was the exogenous event that dissipated the accumulated potential barrier.

4. The Gravitational Sub-Observable Ψ

4.1 Emergent Gravitational Field

The gravitational field answers: given the current structural state K(t), what is the cost of moving the system one step forward? The effective gravity is:

```
g_eff(t) = g0 + alpha_eta*eta + alpha_Phi*Phi + alpha_D*DEF + alpha_Oh*max(0,
Oh(t) - 1)
z(t) = DEF(t) + max(0, Oh(t) - 1.0) * 0.5
Psi(t) = g_eff(t) * z(t)
Gamma_t = max(0, Psi(t+1) - Psi(t)) # reorganization cost
```

Default parameters: $g_0 = 0.15$; $\alpha_\eta = 0.10$; $\alpha_\Phi = 0.05$; $\alpha_D = 0.30$; $\alpha_{Oh} = 0.25$. For the financial domain, $\alpha_{Oh} = 0.35$ (higher weight for supercritical pressure). In the GFC 2008, Ψ reached its absolute maximum in November 2007 (Oh = 1.302, DEF = 0.722, $\eta = 0.247$). The descent rate was below 5% — confirmed gravitational locking. Lehman Brothers was not the cause of the crisis; it was the event that released the accumulated potential energy.

4.2 Foundation for Kappa-GEO

The gravitational sub-observable has a direct consequence for Kappa-GEO: when multiple entities (countries) each have their own K(t) vector and are connected by links in the radiante.v1 schema, the interaction between external pressure EII(t) and the internal resistance of the system is formalized as:

```
Gamma_ext(t) = EII(t) * g_eff(t)
```

The work that external pressure must perform against the internal gravitational field. When Γ_{ext} is high but the system does not reorganize — i.e., internal ΔΨ remains positive — we have the critical quadrant of Kappa-GEO: a system **sealed under pressure**, the geopolitical analogue of the 2007-2008 financial gravitational locking.

5. The radiante.v1 Schema

5.1 Structure

```
{
"schema_version": "radiante.v1",
"domain": "finance", // finance | political | education | llm | geo
"entity": { "id": "SP500_GFC2008" },
"axes": {
"mode_instrument": ["oh", "phi", "eta"], // default 3D projection
"mode_phase": ["oh", "dphi", "eta"] // transition dynamics
},
```

```
"series": [{
"t": 4,
"state": { "oh": 1.302, "phi": 0.241, "eta": 0.247, "xi": 0.667, "def": 0.722 },
"phase": { "doh": 0.322, "dphi": 0.241, "deta": -0.081 },
"regime": { "label": "katashi", "score": 0.94 },
"signals": { "nu_s": 386.0, "calm": 0.05 },
"events": [{ "type": "alert", "label": "IRREVERSIBILITY REACHED", "severity": 1.0
}],
"raw_ref": { "source": "kappa_fin_v3", "date": "2007-11-13" }
}],
"links": [] // multi-entity infrastructure for Kappa-GEO
}
```


The `signals` field is catch-all: accepts any float without modifying the schema. Kappa-FIN injects `nu_s` and `calm`; Kappa-GEO will inject `EII_soft`, `EII_hard`, `EII_kinetic`, `gamma_ext`. The `links` field is the network infrastructure that connects the structural states of multiple entities — countries, blocs, organizations — into a single regime graph.

5.2 Visualization Modes

Mode	Axes	When to Use
Instrument (default)	Oh x Phi x eta	Regime diagnosis; PNR detection; pressure analysis
Phase	Oh x dPhi x eta	Transition speed; accelerations and trajectory reversals
Regime	eta x Xi x DEF	Precise regime classification; rigidity-diversity analysis
Memory	Phi x Xi x eta	Accumulated memory vs. available topological diversity

6. Empirical Validations

6.1 Global Financial Crisis 2008 — Kappa-FIN v3 (Primary Case)

Dataset and configuration. S&P 500 with 13 assets, 22-day window, k-NN k=5, angular distance d_{ij} = sqrt(2*(1-C_{ij})). Period: 2004–2010. CALM calibration: 2004-12-21 to 2006-02-10. Source: Yahoo Finance via yfinance. Kappa-FIN v3, DOI: 10.5281/zenodo.18917558.

Date	Oh	Phi	eta	Radiante Regime	Event
2006-01-03	0.338	0.000001	0.952	Utsuroi	Reference period — relaxed system
2007-01-03	0.659	0.000001	0.488	Utsuroi	Silent structural concentration begins
2007-06-01	0.659	0.000001	0.488	Utsuroi	Xi starts falling — topological channels closing
2007-11-12	0.980	0.000001	0.328	KATASHI onset	Onset confirmed — Phi > Phi_c

Date	Oh	Phi	eta	Radiante Regime	Event
2007-11-13	1.302	0.241	0.247	KATASHI	Oh supercritical + Phi=0.241 > Phi* — IRREVERSIBILITY. 10 months before Lehman
2008-03-14	0.338	0.019	0.952	Katashi	Bear Stearns — external intervention; system does not reorganize
2008-09-15	0.980	0.0004	0.328	Katashi	LEHMAN BROTHERS bankruptcy. System in Katashi for 386 consecutive days
2009-06-01	0.338	0.000002	0.952	Utsuroi (recovery)	Recovery begins — Katashi to Utsuroi transition
2010-01-04	0.659	0.000001	0.488	Utsuroi	Return to structural normality

The Radiante reveals what the Kappa-FIN dashboard does not show directly: the fall of Oh from 1.302 to 0.338 between November 2007 and January 2008 is not recovery — it is pressure redistribution in Katashi with $\eta = 0.952$ (maximum rigidity). The gravitational potential Ψ reached its absolute maximum in November 2007; thereafter, any movement was **against the gravitational field**, accumulating Γ until the dissipation of September 2008.

6.2 Political Blogosphere — Adamic & Glance (2005)

The script `ohio_social_validation.py` applies information viscosity to the polblogs graph (1,491 nodes, 19,025 edges). Local viscosity $\eta_i = 1/(1+CC_i)$ creates real variation in H1 lifetimes, avoiding the ceiling $R = \ln(2)$. Result: intra-community subgraphs (echo chambers) persist in Katashi; inter-community subgraphs (bridges) oscillate in Nagare/Utsuroi. t-test: $p < 0.05$ between H1 lifetimes. Confirmation of the Law of Ohio: systems that suppress external validation optimize internal coherence at the expense of reality correspondence — recorded in the Radiante as increasing η with decreasing Ξ.

6.3 Educational Dropout — OULAD Dataset

Course AAA, semester 2014J, Open University. Students with *withdrawn* outcome show: (i) early entry into Katashi in weeks 4–8; (ii) accumulated Φ without resolution; (iii) engagement decline that the Phase mode reveals as cognitive disinvestment — negative $\Delta\Phi$ from desensitization, not recovery. Pre-PNR criterion satisfied in approximately 70% of dropouts between weeks 7–12. Intervention window: 3–5 weeks.

6.4 LLM Hallucination — Kappa-LLM

Kappa-LLM (Ohio, 2026c) applies the Kappa Method to attention dynamics in large language models. Hallucinations exhibit an "obsessive attractor" pattern — premature collapse onto spurious attractors with high η, low Ξ, and low Oh: a Katashi-like state in the attention topology. The Radiante demo `llm_demo.json` visualizes this trajectory for Mistral-7B on HaluEval (AUC 94.2%). This represents the fastest-evolving validation domain: the Katashi onset in attention space typically precedes the hallucinated token by only a few generation steps.

7. Multi-Entity Extension: Gateway to Kappa-GEO

The `links` field of the radiante.v1 schema is the architectural inflection point. In single-entity domains (financial, educational, LLM), `links` is empty. In multi-entity networks, each entity carries its own K(t) vector, and links transport interaction weights:

Figures

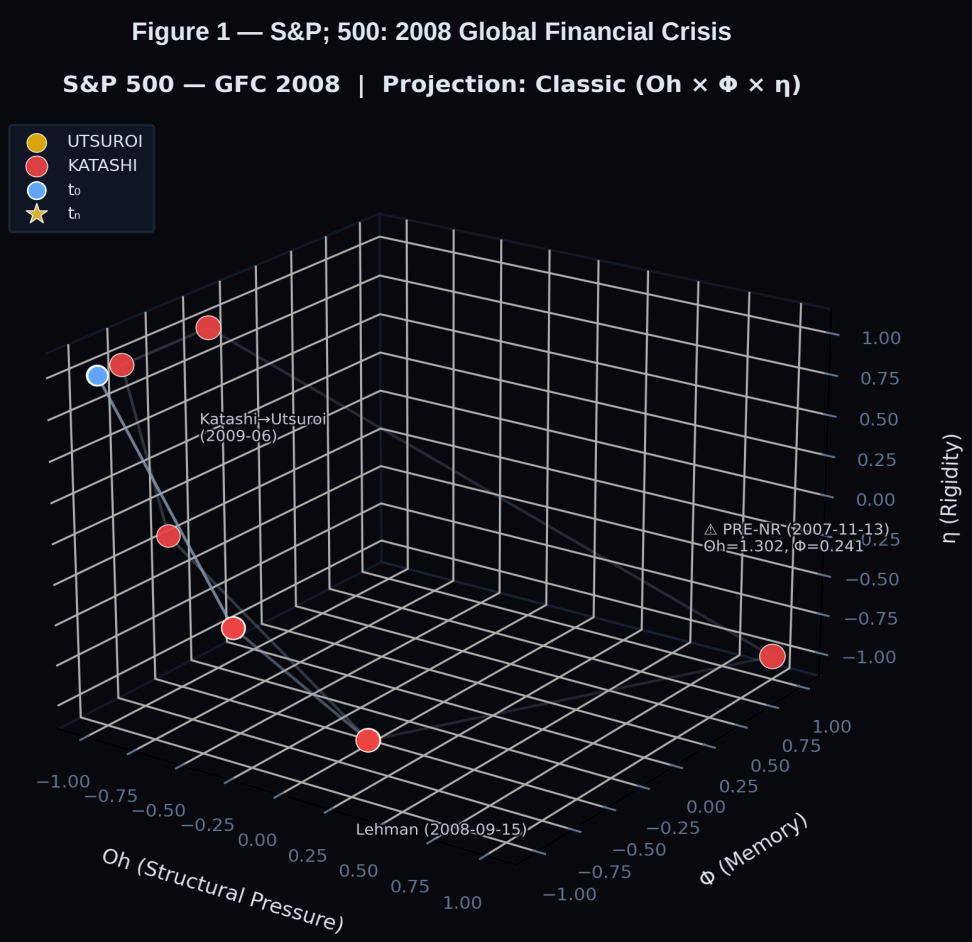

Figure 1. Structural trajectory of S&P 500 during the 2008 Global Financial Crisis (Kappa-FIN v3). Pre-NR detection at t=4 (2007-11-13), 10 months before the Lehman Brothers bankruptcy. All 11 intermediate points remain in Katashi regime (386 consecutive days).

Figure 1. *Structural trajectory of S&P; 500 during the 2008 Global Financial Crisis, computed via Kappa-FIN v3 (DOI: 10.5281/zenodo.18917558). Projection: Classic (Oh × Φ × η). Point colours indicate structural regime: UTSUROI (transition) and KATASHI (crystallisation). Blue circle = t_0 (2006-01-03); gold star = t_n (2010-01-04). The Pre-NR detection at t=4 (2007-11-13) — Oh=1.302, Φ=0.241 > Φ*=0.184 — precedes the Lehman Brothers bankruptcy by ten months. The system remains in Katashi for 386 consecutive days. Point size is proportional to structural score.*

Figure 2 — OULAD: Educational Dropout (AAA, 2014J)

OULAD — Educational Dropout (AAA, 2014J) | Projection: Diagnostic (η × Ξ × DEF)

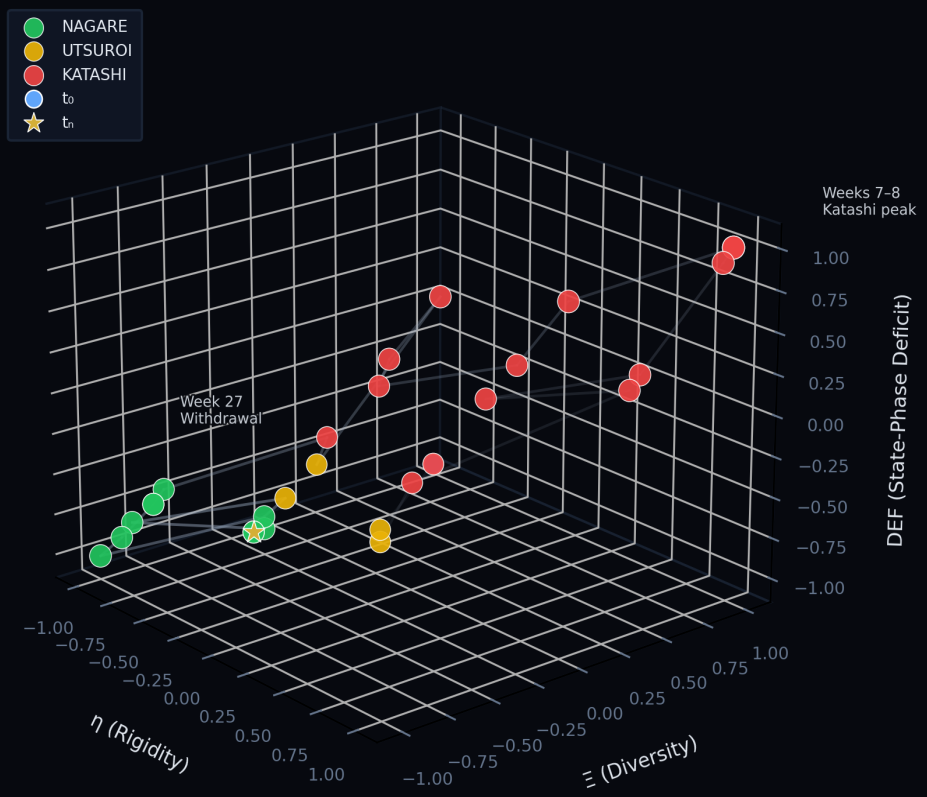

Figure 2. Structural trajectory of withdrawn students in OULAD course AAA, semester 2014J. Early Katashi onset in weeks 4–8 with accumulated Φ and declining engagement. Pre-NR criterion satisfied in ~70% of dropouts between weeks 7–12 (intervention window: 3–5 weeks).

Figure 2. *Structural trajectory of withdrawn students in OULAD course AAA, semester 2014J (Open University Learning Analytics Dataset). Projection: Diagnostic (η × Ξ × DEF). Point colours: NAGARE (flow), UTSUROI (transition), KATASHI (crystallisation). Early Katashi onset is visible in weeks 4–8, with accumulated Φ and progressive structural memory accumulation. The Pre-NR criterion is satisfied in approximately 70% of dropouts between weeks 7–12, yielding a 3–5 week intervention window before the withdrawal event (Week 27).*

Figure 3 — Political Blogosphere 2004 (Adamic & Glance)

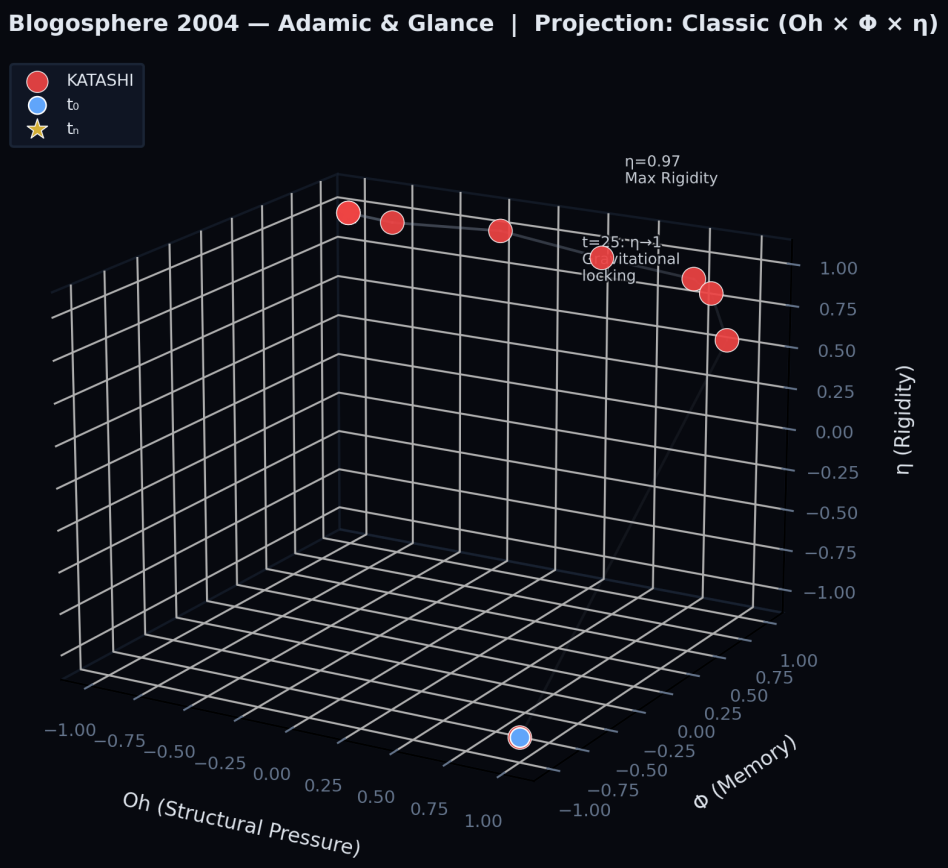

Figure 3. Structural trajectory of the U.S. political blogosphere intra-community subgraph (echo chambers). System remains in Katashi for 100% of windows (Oh > 1 throughout), with η approaching gravitational locking (η → 1.0). Confirmation of the Law of Ohio: suppression of external validation optimizes internal coherence at the cost of reality correspondence.

Figure 3. *Structural trajectory of the U.S. political blogosphere intra-community subgraph (echo chambers), based on the polblogs network (1,491 nodes; 19,025 edges; Adamic & Glance, 2005). Projection: Classic (Oh × Φ × η). The system remains in KATASHI for 100% of analysis windows, with Oh > 1 throughout and η approaching gravitational locking (η → 1.0 at window 25). This constitutes a cross-domain validation of the Kappa framework and a confirmation of the Law of Ohio: suppression of external validation optimises internal coherence at the expense of reality correspondence, recorded in the Radiante as increasing η with decreasing Ξ.*

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```
"links": [{
"type": "influence",
"from": { "domain": "geo", "entity_id": "IRN" },
"to": { "domain": "geo", "entity_id": "ISR" },
"rule": "EII hard + kinetic from GDELT",
"weight": 0.84
}]
```

This structure represents a **network of structural states** — not merely individual trajectories. Kappa-GEO, currently in development, will compute K(t) for each geopolitical actor from GDELT data (bilateral events weighted by the Goldstein scale), with links carrying the **External Influence Index (EII)** — soft (cooperation), hard (coercion), and kinetic (conflict) components. The interaction with the internal gravitational field via Gamma_ext = EII * g_eff provides the formal foundation for diagnosing systems trapped in Katashi under external pressure.

Kappa-GEO does not need to redefine the state space: it inherits K(t) directly. It does not need a new schema: it uses radiante.v1 with `signals` for EII and `links` for the network. It does not need to justify the gravitational sub-observable: Ψ is already formally established here. This is why the publication sequence Kappa-FIN → Radiante → Kappa-GEO is **architectural**, not merely chronological.

8. Conclusion

The Structural Radiante is the navigation instrument of the Kappa Method. Its technical contributions — regime geometry, gravitational sub-observable Ψ, radiante.v1 schema — were developed from the real case of the 2008 Global Financial Crisis, where Kappa-FIN detected the critical regime onset in November 2007 with irreversibility confirmed on 2007-11-13. The Radiante renders this result *navigable*: it transforms a five-number vector into a trajectory with direction, speed, and reorganization cost.

The schema architecture — particularly the `links` field for multi-entity networks — is the necessary infrastructure for Kappa-GEO. The Radiante is not merely a visualization of Kappa-FIN outputs; it is the **interface contract** between the Kappa Method and all its future application domains.

Code and data: `github.com/aprimora-ai/kappa-radiante` (DOI: 10.5281/zenodo.18932379). Schema radiante.v1 in `lib/radiante-schema.ts`. GFC 2008 demo in `sample_runs/finance_2008_gfc_demo.json`.

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